



The p-adic valuation of a linear recurrence

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Let $(u_n)_{n \geq 0}$ be an integer-valued linear recurrence of order $k \geq 1$. Precisely,

$$u_n = a_1 u_{n-1} + \cdots + a_k u_{n-k} \quad \forall n \geq k,$$

for some $a_1, \dots, a_k \in \mathbb{Z}$.

Given a prime number p , I wonder: **How much is known about the p-adic valuation $v_p(u_n)$? Are there some "explicit" formulas?**

- For $k = 1$, we have simply a geometric progression and clearly it holds

$$v_p(u_n) = v_p(a_1)^n + v_p(u_0).$$

- For $k = 2$, the problem was studied (in the more general setting of linear recurrences in the field of p -adic numbers) by Ward [1], who gave formulas for $v_p(u_n)$. However, those formulas rely on the computation of a p -adic number v (see Theorem 10.1) which is essential as much as difficult as the computation of $v_p(u_n)$, so they do not seem to give really useful information on $v_p(u_n)$ (see Vesselin Dimitrov comments).

The particular case of the Fibonacci sequence $(F_n)_{n \geq 0}$ was also studied by Lengyel [2], who gave practical closed expressions for $v_p(F_n)$, in terms of $v_p(n)$, $z(p) := \min\{n > 0 : p \mid F_n\}$, and $e(p) := v_p(F_{z(p)})$.

- For $k \geq 3$ it seems to me that nothing general is known. I found only another article of Lengyel [3] about the 2-adic valuation of the Tribonacci numbers $(T_n)_{n \geq 0}$.

Surely, without loss of generality, it can be assumed that: $(u_n)_{n \geq 0}$ is not degenerate; it has at least one zero modulo p (and that can be effectively checked since $(u_n)_{n \geq 0}$ is periodic modulo p and the period length is less than p^k); $u_0 \equiv 0 \pmod p$, eventually by shifting $(u_n)_{n \geq 0}$.

Thank you in advance for possible ideas and references!

[1] M. Ward. *The linear p-adic recurrence of order two*. *Illinois J. Math.* (6) 40--52, 1962.

[2] T. Lengyel. *The order of the Fibonacci and Lucas numbers*. *The Fibonacci Quarterly*, (33) 234--239, 1995.

[3] *T. Lengyel. The 2-adic Order of the Tribonacci Numbers and the Equation $T_n = m!$. Journal of Integer Sequences Vol. 17, 2014.*

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edited Aug 16, 2015 at 16:24

asked Aug 15, 2015 at 17:37

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2  Surely this is susceptible to rewriting as $M^n u$ where M is the companion matrix for $\lambda^k - \sum a_i \lambda^{k-i}$, and u is any old start-up vector. The characteristic polynomial determines (modulo the start-up) the valuations (this takes some care), but can be done. – [David Handelmann](#) Aug 15, 2015 at 22:35

 "Lifting the exponent" is also a result of this flavor (for $k = 2$): s3.amazonaws.com/aops-cdn.artofproblemsolving.com/resources/... – [Max Alekseyev](#) Aug 15, 2015 at 22:53

2  I do not think that Ward's paper treats a general two-term recurrence. For example, there is no formula for the 3-adic valuation of $5^n - 2$. Proving that $3^n \nmid 5^n - 2$ for all $n > 1$ was once a well known problem whose solution required the 3-adic variant of Gelfond's estimates on linear forms in two logarithms. – [Vesselin Dimitrov](#) Aug 16, 2015 at 7:14 

 In general, diophantine approximations (Schmidt's theorem) yield a good, but ineffective upper bound on the p -adic valuation of u_n . – [Vesselin Dimitrov](#) Aug 16, 2015 at 7:15

1  This is correct. With the normalization $5 \cdot 5^n - 2 \cdot 1^n$, the example I gave falls into Theorem 10.1 in Ward. But this is just a reformulation of the essential problem, in terms of how well can the 3-adic integer ν in that theorem be 3-adically approximated by a rational integer n . This is a quintessential problem in diophantine approximations, solved by the Gelfond-Baker results of logarithmic linear forms. – [Vesselin Dimitrov](#) Aug 16, 2015 at 9:48 

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2 Answers

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Actually, the binary linear recurrence case is pretty precise, especially if $p \geq 3$ and you're working over $\mathbb{Q}$, and not over a field where p is ramified. Let $r(p)$ denote the rank of apparition, which is the smallest r such that $p \mid a_r$. Then if $p \geq 3$, we have

$$\text{ord}_p(a_n) = \begin{cases} 0 & \text{if } r(p) \nmid n, \\ \text{ord}_p(a_{r(p)}) + \text{ord}_p(n/r(p)) & \text{if } r(p) \mid n. \end{cases}$$



In the literature there are sources which state this as a theorem for all primes in all number fields, but it's not quite right for $p = 2$ over $\mathbb{Q}$, nor for larger p if p is ramified. The best source that I know for the general result is:

Stange, Katherine E., Integral points on elliptic curves and explicit valuations of division polynomials. *Canad. J. Math.* **68** (2016), no. 5, 1120-1158. (MR3536930)

<http://arxiv.org/abs/1108.3051>

The case of $p \geq 3$ over $\mathbb{Q}$ is a fairly easy exercise once one understands that underlying these binary linear recurrences (and also underlying elliptic divisibility sequences) is a one-dimensional algebraic group, and the desired result follows from a calculation in the formal group.

On the other hand, it is a very hard (pretty much open) problem to determine if there are infinitely many n such that $\text{ord}_p(a_{r(p)}) \geq 2$, or even that there are infinitely many n such that $\text{ord}_p(a_{r(p)}) = 1$, although presumably the latter occurs for almost all n . Both problems are open even for $a_n = 2^n - 1$.

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edited Aug 20, 2017 at 14:52

answered Nov 1, 2015 at 20:15



Joe Silverman

- 3 ▲ Are you considering an arbitrary binary linear recurrence? This applies to $a_n = q^n - 1$, and if I understand correctly, a similar result holds for elliptic divisibility sequences. But take $p = 3$ and $a_n = 5^n - 2$. The rank of apparition is $n = 1$, with corresponding valuation 1, but the 3-adic valuation of a_n is zero for all even n . Moreover, for $n = 5$ we have $3^2 \mid 5^5 - 2 = a_5$, and then for $n = 11$ we have $3^3 \mid 5^{11} - 2 = a_{11}$. So, in this example, the 3-adic valuation is rather subtle; indeed it depends on the rational approximations to a 3-adic log. – Vesselin Dimitrov Nov 1, 2015 at 22:27
- ▲ @Joe Silverman: I am confused by your last paragraph because the (in)equalities have no dependency on n . Is it about an infinite number of primes p ? – Luc Guyot Sep 16, 2016 at 19:02
- 1 ▲ @LucGuyot Yes, sorry, in both cases "infinitely many n " should be "infinitely many p ", and ditto "almost all n " should be "almost all p ". – Joe Silverman Sep 16, 2016 at 19:39

Regarding p -adic valuation of Lucas sequences, a quite precise result is given in [1].

3

Theorem. Let $(u_n)_{n \geq 0}$ be a nondegenerate Lucas sequence with $u_0 = 0$, $u_1 = 1$, and $u_{n+2} = au_{n+1} + bu_n$ for all $n \geq 0$, where a and b are two integers. Furthermore, let p be a prime number not dividing b .

Then for any positive integer n we have

$$v_p(u_n) = \begin{cases} v_p(n) + v_p(u_p) - 1 & \text{if } p \mid \Delta, p \mid n, \\ 0 & \text{if } p \mid \Delta, p \nmid n, \\ v_p(n) + v_p(u_{p\tau(p)}) - 1 & \text{if } p \nmid \Delta, \tau(p) \mid n, p \mid n, \\ v_p(u_{\tau(p)}) & \text{if } p \nmid \Delta, \tau(p) \mid n, p \nmid n, \\ 0 & \text{if } p \nmid \Delta, \tau(p) \nmid n, \end{cases}$$

where $\Delta := a^2 + 4b$ and $\tau(p)$ is the rank of apparition of p in $(u_n)_{n \geq 0}$, i.e., the least positive integer m such that $p \mid u_m$. Moreover, if $p \geq 3$ then

$$v_p(u_n) = \begin{cases} v_p(n) + v_p(u_p) - 1 & \text{if } p \mid \Delta, p \mid n, \\ 0 & \text{if } p \mid \Delta, p \nmid n, \\ v_p(n) + v_p(u_{\tau(p)}) & \text{if } p \nmid \Delta, \tau(p) \mid n, \\ 0 & \text{if } p \nmid \Delta, \tau(p) \nmid n, \end{cases}$$

while if $p \geq 5$ then

$$v_p(u_n) = \begin{cases} v_p(n) & \text{if } p \mid \Delta, \\ v_p(n) + v_p(u_{\tau(p)}) & \text{if } p \nmid \Delta, \tau(p) \mid n, \\ 0 & \text{if } p \nmid \Delta, \tau(p) \nmid n. \end{cases}$$

Actually, in [1] the theorem is stated for a and b relatively prime. However, as explained in [2], the result holds even if a and b are not coprime.

[1] C. Sanna, *The p-Adic Valuation of Lucas Sequences*, Fibonacci Quart. **54** (2016), no. 2, 118–124. (Free preprint: https://www.researchgate.net/publication/304251918_The_p-adic_valuation_of_Lucas_sequences)

[2] N. Murru, C. Sanna, *On the k-regularity of the k-adic valuation of Lucas sequences* <http://arxiv.org/abs/1603.09310>

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edited Jun 25, 2016 at 7:41

answered Jun 4, 2016 at 13:40

user40023